Function Evaluation vs Dot Product

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Dot Products

Consider the following equation

$$f(x) = x^2 + 2x + 4$$

A simple evaluation gives the following result

Let $x = 2$, then

$$\begin{aligned} f(2) &= 2^2 + 2(2) + 4 \\ &= 12 \end{aligned}$$

However, suppose we want to augment the above result in a dot product form (for Machine Learning purposes)

We know that

$$\bar{x}^T y = \sum_i x_i y_i$$

Let us model the above equation in a vector form

$$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix}, x = \begin{bmatrix} x^2 \\ x \\ 1 \end{bmatrix}$$

, and $x = 2$

$$\begin{aligned} \bar{\theta}^T x = \sum_i \theta_i x_i &= [\theta_0 \quad \theta_1 \quad \dots \quad \theta_n] \begin{bmatrix} x^n \\ x^{n-1} \\ \vdots \\ 1 \end{bmatrix} \\ &= [1 \quad 2 \quad 4] \begin{bmatrix} (2)^2 \\ (2) \\ 1 \end{bmatrix} \\ &= [1 \quad 2 \quad 4] \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix} \\ &= 1(4) + 2(2) + 4 = 12 \end{aligned}$$